

ZAIDH RIZME

MECHATRONICS & ROBOTICS ENGINEER

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SUMMARY

Mechatronics Engineer with hands-on experience in PCB design, 3D CAD modelling, embedded systems, and industrial automation. Capable of developing electronic schematics and PCB layouts, creating SolidWorks parts and mechanical assemblies, and integrating sensors, microcontrollers, PLCs, and vision systems. Experienced in C/C++, MATLAB, prototyping, testing, and technical documentation.

PROFESSIONAL EXPERIENCE

Autonomation Engineer | MAS Capital - Intimates Division (Pvt) Ltd. Jul 2025 - Feb 2026

- Implemented autonomation systems to reduce downtime and defects in manufacturing lines.
- Integrated sensors, PLCs, and vision systems for process monitoring and quality control.
- Designed 3D CAD models and mechanical assemblies for autonomation systems, translating functional requirements into practical component designs.
- Assisted in testing, troubleshooting, and documentation of autonomation solutions.

PROJECTS

ToF Based SLAM and Autonomous Mobile Robot Dec 2025 - Present

University of Wolverhampton | Final Year Project

- Development phase of a ToF-based SLAM system for autonomous localization and mapping.
- Implement sensor fusion using IMU and wheel encoder data for pose estimation.
- Designed embedded real-time navigation and motion control on ESP32.
- Developed MATLAB tools for mapping, visualization, and performance analysis.
- Integrated and validated robotic sensing, localization, and navigation subsystems.

Automated Modified Iron July 2025 - Dec 2025

MAS Capital - Intimates Division (Pvt) Ltd

- Developed an industrial iron with relay, timer, and temperature control for precise heating.
- Integrated pneumatic cylinder and movable arms to automate garment pre-heating before pressing.
- Improved efficiency, consistency, and reduced manual effort in branded apparel manufacturing.
- Conducted testing, troubleshooting, and optimization to ensure reliable production use.

Automated Oil Level Monitoring System for Industrial Machinery Oct 2024 - April 2025

Brother Industries Pvt Ltd | University of Wolverhampton

- Designed the electronic schematic and PCB for an ATmega328P/MS5837 oil-level monitoring system.
- Developed C firmware to process data, trigger alerts, and ensure reliable industrial performance.

Wearable Active Posture Corrector Device (Research) Jan 2024 - April 2024

OREL Corporation Pvt Ltd | University of Wolverhampton

- Investigated the use of neodymium magnets and Hall effect sensors to detect upper body posture deviations with high sensitivity.

CORE ENGINEERING SKILLS

PCB Design

Custom PCB for the ToF robot with a 4-layer stack-up and copper-zone filling.

- Schematic capture, PCB layout, sensor interfacing and embedded-system integration.

3D CAD & Mechanical Design

Part design, high-level assemblies, simulations, BOM creation and technical drawings.

- Applied to automation equipment | robotics | manufacturable mechanical solutions.

PROGRAMMING & AUTOMATION

- Programming: MATLAB, C++, Python, C, HTML and CSS.
- Applications: ToF robot, oil-monitoring system and MAS Intimates automation projects.
- Embedded systems: ESP32, ATmega328P, Arduino, sensors and actuators.
- Industrial automation: PLCs, vision systems, pneumatics, testing and troubleshooting.

ADDITIONAL EXPERIENCE

Academic Assistant | LivePhysics (Pvt) Ltd.

July 2024 - July 2025

- Preparing and documentation of GCE Advanced Level Physics Papers, Marking Schemes, Lesson presentations.
 - Coordinate with staff for scheduling and material planning.
- Skills gained: Excelled in MS Office documentation, Analytical Thinking, Technical Communication and Collaboration, Teamwork.

Course Instructor | NEOGRAENIUM

Jan 2023 - Sep 2024

- Developed and implemented dynamic curriculum for LEGO robotics and Micro:Bit device.

EDUCATION

Bachelor of Engineering (Hons) in Mechatronics Engineering

Sep 2023 - Sep 2026

University of Wolverhampton | CINEC Campus Pvt LTD

- Gained hands on experience in embedded systems and automation through practical labs and individual projects.

GCE Advanced Level (Physical Science Stream)

Sep 2020 - Dec 2022

Isipathana College, Colombo 05

- Combined Mathematics - B | Chemistry - B | Physics - C

CERTIFICATIONS

- Diploma in Information and Technology (Distinction)
- Sri Lankan Mathematics Olympiad (Distinction)
- Specialized in 3D CAD Modelling using Solidworks (Coursera)
- C Programming / C++ Programming at Udemy

REFEREES

1. **Name:** Randeera Liyanage, Msc

Position: Lecturer in Mechatronics Engineering | Secretary - CIRS 2025 | Chairman - Industry Consultative Committee

Organization: Faculty of Engineering and Technology at CINEC Campus (Pvt) Ltd.

Email: randeera.liyanage@cinec.edu

Contact Number: +94 77 714 2029

Relationship: Lecturer | Counsellor of IEEE Student Branch CINEC Campus 2025

2. **Name:** Dr. Sanjeewa Dharmawardhana, Ph.D.

Position: Educator of Physics Subject for GCE A/L Students | Educational Consulting

Organization: LivePhysics (Pvt) Ltd.

Email: sanjeewadharmawardhana@gmail.com

Contact Number: +94 77 022 0222

Relationship: Founder | Supervisor at LivePhysics (Pvt) Ltd.